

Grain direction, edging and allowances

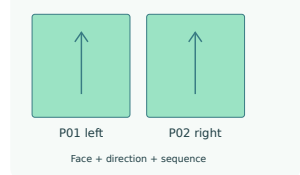
A finished part dimension may differ from its core cutting dimension. Specify grain, edge treatment and measurement stage, then calculate allowances from the defined construction.

What you will accomplish

- Specify grain direction for each part
- Calculate core size from defined edges
- Document allowance and operation

Specify grain direction for each part

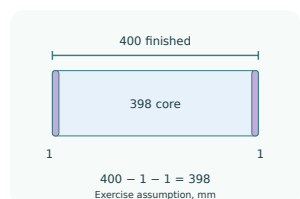
Place a grain arrow on the identified face. For adjacent fronts, state orientation and sequence when a continuous appearance is required. Rotating a rectangle in a cutting layout may save material but change grain. Do not assume the first scheduled dimension defines grain direction; record it separately.



A grain arrow is independent of dimension order.

Calculate core size from defined edges

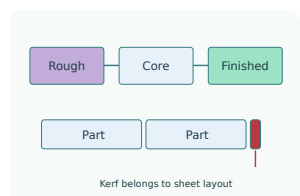
In this declared exercise, finished width is 400 mm and opposite edges each receive an assumed 1 mm band, with no other additions in that direction. Core width is $400 - 1 - 1 = 398$ mm; one banded edge would give 399 mm. Label stage, edges and system. This geometric assumption is not a universal product recipe.



Exercise only: 400 finished
= 1 edge + 398 core + 1 edge.

Document allowance and operation

Distinguish rough preparation stock from edge deductions and assembly clearances. Saw kerf affects sheet layout, not an automatic addition to each finished part. Record finished, core and rough-stage dimensions with reasons. A 402 mm schedule entry against a 400 mm drawing requires clarification of stage or revision, rather than an unexplained correction.



Every dimensional difference needs a stage and reason.

Worked case and decision

Both exercise fronts finish at 400 mm width. P01 has two 1 mm opposite bands, giving a 398 mm core; P02 has one 1 mm band, giving 399 mm. Mark E1/E2, grain and measurement stage. A bare 400 mm entry does not establish the required cut size.

Decision: schedule 398 and 399 mm as core sizes under the documented exercise assumptions.

Check understanding

For a 400 mm finished width with two 1 mm bands, what is the exercise core width?

- 398 mm
- 400 mm
- 402 mm

Correct: 398 mm. Under the stated assumption: $400 - 1 - 1 = 398$ mm.

Lesson questions

Should kerf be added to every finished part?

No. Account for it in the sheet layout separately from finished part dimensions.

Practice

Prepare P01/P02 rows, a face sketch with grain and edge IDs, and a table of finished/core dimensions with reasons.

Face and edge drawing

Measurement-stage table

Calculation assumptions

Review your work against the criteria. Saving it does not constitute technical project approval.

- Deductions match identified edges
- Grain stated for each face
- Every allowance has a reason